

Examiner reports

Q1.

Part (a) was done well by the majority of students. Most knew how to set up the initial differential equation and those who did generally scored full marks. There were a few numerical slips seen and some students did not complete the argument to show the required result.

A high proportion of students made good progress on part (b) with over half of students scoring 2 or 3 marks. Having correctly integrated, many students gave no consideration to the constant of integration and, although it was zero, its value needed to be found to correctly show the given result.

Part (c)(i) was well attempted and saw the full range of marks awarded. Over half of the students scored at least 4 marks with 6/6 being the most frequent score awarded. Many students realised that simultaneous equations had to be solved and usually eliminated a variable to form a correct equation. Students often made algebraic slips when solving their equation and a more efficient use of allowed calculator functions would have helped here.

In part (c)(ii) many students made some good criticisms of the equations they had found, but did not link these to the model used by the trader, which was the rate of sales is

proportional to $\frac{8-t}{x}$.

Q4.

Since the form of the required differential equation was given, students could decide the values of a and k quickly and then give their differential equation very quickly. Although some did this, many did some working before often coming to wrong conclusions about the values of a and k . Some calculated second derivatives whilst some tried to integrate their differential equation, all quite unnecessarily. It was disappointing that many did not realise that a has to be 1.3, corresponding with the maximum value of the cosine term

being 1. Most students did write down $\frac{dh}{dt}$ and about 80% of the entry scored 1 or more of the 3 marks, but only about 5% got the question fully correct with relatively few students getting the value of k correct; the π was often missing from those who had shown some

understanding, giving $k = \frac{1}{6}$ or $k = \frac{1}{12}$.

Q5.

For part (a), some student wrote down the correct differential equation apparently fully understanding all the information given and interpreting it correctly. However, all sorts of errors abounded in other attempts, some not even involving a derivative, and some with derivatives in x and y . Some attempts bore little or no relation to the information given, although credit was given if $\pm (2 - h)$ appeared in an otherwise incorrect expression. Many had a spurious t and l or h , either as a multiple or power, and the k appeared in a variety of places. Some students did not even form an equation, leaving a proportionality sign in their answer.

In part (b), most students knew they were expected to separate the variables and did it correctly, although there were some notation errors in the positioning of dx , at the front rather than the rear of the integrand. Those who failed to separate the variables, just produced nonsense.

Most students integrated $k dt$ successfully, but relatively few made good progress with the integral in x . No guidance was given, but students split fairly evenly between attempts by parts and by substitution, with similar success, or lack of it, via either method. Many students just integrated each term in the product; presumably this was an attempt at parts. Those who showed the parts and how the integral might be achieved scored a method

mark, but many could not integrate $\sqrt{(2x-1)}$ correctly, making an error in the required coefficient.

Those who used substitution usually chose to put $u = 2x - 1$ but did not always change their whole integral to be in terms of u ; a missing du was penalised for the method mark. Some with a correct first stage then did not multiply out to obtain an integrable form, but just integrated each term in the product. All students, no matter what their attempt at the integral, could obtain a method mark if they included a constant and tried to find it using the given initial conditions. Those few who did integrate and find the constant correctly, usually went on to calculate the final requested value of t correctly.

Q6.

In part (a) most candidates at least used $\frac{dA}{dt}$ in their attempt at a differential equation although

some did use other letters. Very few equated to a constant of either sign. The term “constant rate” as given in the question just wasn’t understood, with most taking the rate of change to be proportional to the surface area, or to time or functions of these such as an exponential function. Only about 5% of the candidates set up the correct equation, which meant they could achieve little in part (b)(i).

In part (b)(i) those who had set up a correct equation in either $\pm k$ usually completed correctly, using integration and the conditions given. Some did attempt to integrate their differential equation, and this was given credit; many did get as far as a value for k , in which algebraic and numerical errors were condoned for the method mark, but somewhat less than 15% of the candidates did get this far.

In (b)(ii) most candidates were able to interpret the given solution in part (b)(i) to answer this correctly with over 60% of the candidates gaining this 1 mark.

Q7.

Most candidates answered part (a)(i) correctly, although a surprising number seemed to think that $A \times 0 = A$.

Most candidates did part (a)(ii) convincingly, usually choosing to evaluate the exponential expression as 0.95 ... and solving for A to demonstrate its value is 60 to two significant figures. Those who did not actually solve for A or did not show A as 59.9... were penalised.

Most candidates started part (b)(i) well, showing they had interpreted the information

given correctly. However many could not manipulate their resulting $\ln$ expression into the form required; many, either not knowing what was meant by “ a and b are integers” or not taking notice of it, left their result in fractions. Many candidates ‘lost’ a minus sign during their working whilst some attempted to take the $\ln$ of a negative number.

In part (b)(ii), a few candidates were able to derive the result clearly and efficiently; others managed it after some working that was somewhat difficult to follow whilst others made an error such as dropping a minus sign but still claimed they had obtained the result.

However, many candidates gave up after finding $\frac{dh}{dt}$. A few candidates attempted to integrate the given differential equation, some successfully recovering the given expression for A , which was both acceptable and impressive.

Although many candidates had part (b)(iii) correct, many errors were also seen, quite a

common one being $4 \times 2 = \frac{1}{2}$. The other common error was to substitute $h = 13$, rather

than $\frac{dh}{dt} = 13$. It was notable that many candidates attempted part (b)(iii) before returning to try and sort out part (b)(ii), which was sensible.

Q8.

- (a) Attempts at this question varied from no attempt, through superficial attempts to some fully correct solutions. Most could separate the variables correctly, although a few did attempt “integrals” of $f(x)dt$. Some candidates inverted the given differential equation and so could attempt direct integration, which was a quite acceptable method. Most integrated kdt correctly, but the integral of $(x+1)^{-\frac{1}{2}}$ defeated many. Often there was an error in the power, before or after integration, and also in the coefficient. However, most candidates did include a constant and attempted to find it from the given initial conditions, for which they gained credit. However, some did some algebraic manipulation before adding a constant, and so could gain no credit, as this was then incorrect. Many candidates who had correctly integrated the expressions failed to gain the last mark, which was for giving the answer in the form $x = f(t)$, because they couldn’t square their expression correctly, often omitting the “middle” term.
- (b) Candidates who had interpreted the context correctly could gain at least one mark here by substituting $t = 60$ into their result from part (a) and solving for x . However, many confused the given $t = 60$ with the given initial condition $x = 80$. Relatively few correct answers were seen, with many candidates obtaining results of over 100% or indeed negative results. These were not apparently queried, with little evidence of checking for errors.
- (c) (i) Very few candidates wrote down a correct differential equation. Some who had seemingly understood the context, wrote down the correct expression in A , but did not include the constant of proportionality. Many candidates mixed up the variables with x , and less so y , appearing, although most had a “differential equation” of some sort. Some candidates even included units in their differential equation, which was condoned if at the end of a correct expression.

- (ii) For many candidates this was the only part of this question, they attempted. Candidates who had understood the context and started with $A = 4.5$ could mostly solve the equation for t successfully. The relatively rare error was in the algebraic manipulation or in attempts to take logarithms, such as $\log(1 + 4e^{-0.09t}) = \log 1 - 0.09 \log 4t$ or similar. Some made an error in miscopying -0.09 .

Those candidates who commonly started with $A = 50$ or $A = 0.5$ encountered the need to take the logarithm of a negative number in their working or obtained a negative value for t , which most then chose to ignore. They gained no credit. Some did realise their error in interpreting the context, and reverted to the correct value for A .

Q9.

Many candidates seemed to have little experience of formulating a differential equation as often a derivative was missing from their answer to part (a)(i). Attempts at a derivative often involved variables other than the given x and t . Some candidates interpreted the question as 'find x in terms of t ', and many expressions involving an exponential function were seen. Some candidates, who essentially had a correct differential equation, also thought they were required to solve it. Few candidates actually responded to the word 'decreasing' in the question, and many had a minus sign missing from an otherwise correct expression.

The given values of 500 and 20000 were often misinterpreted in part (a)(ii); the 500 often being taken to mean time. However, many candidates did indicate they had understood 500 to be a rate and 20000 the current value of x , but often also had a spurious t or another term in x in their differential equation. Some had a correct value of k , but negative, not apparently noting $k > 0$ had been given in the question. Relatively few fully correct answers to part (a) were seen.

Most candidates found the value of A correctly in part (b)(i), the common error being an apparent misinterpretation of the given information and finding A to be 700.

Part (b)(i) of the question was generally done well with many fully correct answers, and very few solutions by a trial and improvement approach. The majority of candidates could manipulate the expression and take logarithms correctly. Some algebraic errors were seen; in particular those who divided 1900 by 2000, rather than subtracting it, gained no further credit. Some first subtracted the 2000, and then tried to take logarithms of the expression as it stood, usually with an error. Some candidates incorrectly used base 10 logarithms. In completing the question most candidates rounded down their result to give the correct year. Some, who had made an error, did not round down and so had their year incorrect. Some candidates misinterpreted the last request in the question giving their answer as the 51st or 52nd year, rather than 2059.

Q10.

Most candidates struggled with part (a) of this question, indicating inexperience, or non understanding, of the formulation of a differential equation from a context situation.

Part (a) Despite the question explicitly requesting a differential equation for $\frac{dx}{dt}$ many

candidates just wrote down an expression for x .

A commonly seen differential equation was $\frac{dx}{dt} = kx$ which candidates then proceeded to solve, despite the question stating that a solution of the differential equation was not required. Of those who could translate the given information into correct symbols, a symbol t for time, often found its way into their expression, and then often in an exponential form. Many attempts using the exponential function were seen, suggesting many candidates associating anything to do with differential equations with exponential forms. Those candidates who did give a correct differential equation in part (i) usually found a correct value for the constant of proportionality k . However, there were also many attempts from a partially correct part (i) where the given values of 1000 and 200 were confused, the latter often being taken as a value of time. Other candidates put forward the incorrect argument that the population would be equal to 1200 at time $t = 1$.

Part (b)(i) Virtually all candidates scored at least one mark here; the common error being to round off 5.545 to 5.6 or not to round it at all.

Part (b)(ii) Many candidates solved this equation correctly, although it was not always clear just how they had done it. Most candidates used the technique as per the mark

scheme, although some worked successfully from $e^{30} = \left(\frac{4x}{5000-x}\right)^4$. Of those who went awry, the error usually involved an incorrect manipulation of the logarithm expression, including ignoring the $\ln$ altogether, or stating $4\ln 4 = \ln 16$. Some started correctly with $7.5 = \ln 4x - \ln (5000 - x)$ but obtained $e^{75} = 4x - (5000 - x)$. Others made relatively simple errors in signs or coefficients in their manipulation towards a solution.

Q11.

This question was usually answered well. However some students forgot the minus sign in the initial equation $12 \frac{dv}{dt} = -4v^{\frac{1}{3}}$ whilst others had difficulty with the evaluation of $\int v^{-\frac{1}{3}} dv$.

Most students attempted to find the '+c' term and obtained the printed result. Nearly all students correctly found the time when the particle came to rest.

©2025 Exam Papers Practice. All rights reserved.

Q12.

Most students answered part (a) correctly but some did not replace their a by $\frac{dv}{dt}$. In part

(b) students were expected to use $\int \frac{1}{100-v} dv = - \int \frac{1}{40} dt$; some incorrectly split $\int \frac{1}{100-v} dv$ into $\int \frac{1}{100} - \frac{1}{v} dv$, whilst others incorrectly used $\int (100-v) dv = \int \frac{1}{40} dt$.

A common error was in forgetting the minus sign in the evaluation of $\int \frac{1}{100-v} dv$. Most students attempted to find the '+ c' term in the solution of the differential equation.

Q13.

This question was answered well by most candidates. However, in part (a), a few

candidates did not show the required step $0.4 \frac{dv}{dt} = 2 - 4v$; others lost and recreated the minus signs, eg $\frac{dv}{dt} = 5 - 10v = -10(0.5 + v) = -10(v - 0.5)$, which was penalised. In part (b), the majority of candidates used $\int \frac{1}{v-0.5} dv = -\int 10 dt$ and then solved this by integrating both sides.

The integration $\int \frac{1}{v-0.5} dv$ caused some candidates difficulty, finding terms such as $\frac{1}{\frac{1}{2}v^2 - 0.5v}$ etc. The use of logarithms was often incorrect, with $\ln(v - 0.5) = -10t + \ln 0.5$ commonly becoming $v - 0.5 = e^{-10t} + 0.5$

Q14.

Most candidates answered part (a) correctly but some just wrote down the printed result.

In part (a) some candidates started with $m \frac{dv}{dt} = -9.8v$ but found that this did not give the printed result, and just invented a term in g . This was not accepted. Some were penalised for ignoring the m term or for changing signs.

In part (b) candidates were expected to use $\int \frac{1}{v-5} dv = -1.96t$; some split $\int \frac{1}{v-5} dv$ into $\int \frac{1}{v} - \frac{1}{5} dv$. However there were many good answers to this part with both the integrations required being successfully completed. The difficulty which some candidates had was in the simplification needed to convert $\ln(v - 5) = -1.96t + \ln 2$ into $v - 5 = 2e^{-1.96t}$ which often became $v - 5 = e^{-1.96t} + 2$

Q15.

Most candidates answered part (a)(i) correctly but some just wrote down the printed result.

In part (a)(ii), candidates needed to start from $F = ma$ and show the cancellation of the mass (65). Some were penalised for ignoring the m term or ignoring the minus sign.

In part (b), candidates were asked to show that $\int \frac{1}{v-2.45} dv = -\int 4 dt$ many simply used this equation and were penalised. There were, however, many good answers to this part, with both the integrations required being successfully completed. The difficulty which candidates had was in the simplification needed to convert $\ln(v - 2.45) = -4t + \ln 17.15$ into $v - 2.45 = 17.15e^{-4t}$, which often became $v - 2.45 = 17.15 + e^{-4t}$.

Q16.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were penalised for ignoring the m term or ignoring the minus sign. There were many good

answers to part (b), although a significant number of candidates used 'inventive' algebra. A few candidates who integrated to find the equation in v and t forgot the $+c$ term.

Q17.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were penalised for ignoring the m term or ignoring the minus sign. There were many good answers to part (b), although some used inventive algebra. The solution to part (c) was usually correct, although candidates often forgot to obtain ± 1 , for the square root of v , when they found the square-root of the equation obtained in part (b).

In part (d) candidates who integrated v to find the distance travelled often forgot the $+d$ term which was required to be shown to be equal to zero.

Those who used integration with limits commonly used $\int_{16}^1 (4 - 0.1t)^2 dt$ forgetting that the integral was with respect to t so that $\int_0^{30} (4 - 0.1t)^2 dt$ was required. A considerable number of candidates, in part (d), tried to use equations for motion with constant acceleration.

Q18.

The majority of candidates used $\frac{dv}{dt} = -\frac{\lambda}{v^4}$ and then solved this by using $\int v^4 dv = \int -\lambda dt$. Some used $a = \frac{dv}{dt}$, together with $v = u + at$, so that $v = u - \frac{\lambda}{v^4}t$ was often seen.

Many candidates found $\frac{4}{5}v^{\frac{5}{4}} = -\lambda t + \frac{4}{5}v^{\frac{5}{4}}$, but the solution for v was not often found correctly.

A common result was " $v^{\frac{5}{4}} = u^{\frac{5}{4}} - \frac{5}{4}\lambda t \Rightarrow v = u - \left(\frac{5}{4}\lambda t\right)^{\frac{4}{5}}$ ".

Q19.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were

penalised for starting with $\frac{dv}{dt} = -\frac{0.08v^2}{0.05}$. There were many good answers to part (b),

with very few candidates being unable to obtain $v = \frac{15}{5 + 24t}$ from $\frac{1}{v} = 1.6t + \frac{1}{3}$.

Q20.

The answer printed in part (a) enabled virtually all candidates to create their solution. However, a considerable minority were not convincing. Part (b) was well answered, with most candidates appreciating that the differential equation needed to be solved by

separating variables. Some found the resulting integral, $\int v^{-\frac{3}{2}} dv$, challenging, and the $+c$

term was often omitted. In part (c), rather than using $\frac{2}{\sqrt{v}} = 2t + \frac{2}{3}$ to obtain t when

$v = 4$, many used the last result, $v = \frac{36}{(2 + 3t)^2}$ when $v = 4$, and forgot to eliminate the

negative root when calculating the square root of $4 = \frac{36}{(2 + 3t)^2}$.

Q21.

Parts (a), (b) and (c) were answered well. In part (d), few candidates used the fact that the work done by the force was equal to the change in kinetic energy. Most used complicated methods which rarely gave an appropriate answer.

Q22.

Virtually all candidates obtained $\frac{dv}{dt} = -0.05v$; only a few ignored the required step $m \frac{dv}{dt}$

$= -0.05mv$. The equation $\int \frac{dv}{v} = -\int 0.05 dt$ was a necessary step which needed to be seen in part (b). Candidates knew roughly how to obtain $v = 20e^{-0.05t}$. Unfortunately, too often algebraic skills were not sufficient and the equation

In $v = -0.05 + c$ regularly became $v = e^{-0.05t} + c$ before becoming $v = Ae^{-0.05t}$. This and similar errors were not condoned.

Q23.

A significant number of candidates created $\frac{dv}{dt} = -\lambda v$ without proper justification. The equation $-\lambda mv = ma$ was a necessary step which needed to be seen. Even more knew roughly how to obtain $v = Ue^{-\lambda t}$, but too often algebraic skills were not sufficient and In $v = -\lambda t + \ln u$ regularly became $v = e^{-\lambda t} + U$ before becoming $v = Ue^{-\lambda t}$. This and similar errors were not condoned.

Q24.

This question required the use of a differential equation, but some candidates tried to use constant acceleration equations. There were a number of good, complete solutions. Some candidates were on the right track with their solutions, but made algebraic or integration errors. A few candidates omitted the negative sign when they started the question and ended up with a final answer which simply had a missing negative sign.

Q25.

The responses to the modelling in parts (a) and (b) were very varied. Only a small number of candidates gained all three marks, but most gained at least one mark. Some candidates made statements that contradicted the information in the stem of the question, such as “the resistance force will be constant”. Part (c) tended to be done very well or very badly. There was some use of constant acceleration equations, but this was not very common. The biggest issue was the omission of a negative sign in the formation of the differential equation, which then continued throughout the solution. This error was quite common. Other problems were caused by incorrect algebraic manipulation, particularly with logarithms, or integration.

Q26.

Part (a) of this question was often done well. Many candidates seemed to take the question in their stride, but others made limited progress or did not know how to approach the question. There were some who lost accuracy marks. The four main causes of this were;

- Omission of the negative sign in the differential equation,
- Algebraic errors,
- Incorrect integration,
- Not finding the value of the constant of integration.

Part (b) was done very well, with many candidates gaining the available mark without doing part (a).

Q27.

Many candidates produced fully correct and succinct solutions here. Others barely scored any marks, as they seemed unaware of the variable nature of the acceleration and applied the equations of constant acceleration. A few started correctly, but made errors in separating variables prior to integration.

Q29.

Part (a) was done very well, and many of the candidates seemed to have benefited from the printed answer. Part (b) however, produced a wide variety of responses. There was very little use of the constant acceleration equations in this question, which was encouraging. The main problems that the candidates encountered were with the integration. Errors such as omitting a negative sign or incorrectly splitting the $1/50 - v$ into two fractions were the most common errors.

Q30.

There were a number of good solutions to part (a). The main errors were due to

missing negative signs in the differential equation, the use of $v \frac{dv}{dx}$ instead of $\frac{dv}{dt}$ or algebraic errors in the working. Very few candidates tried to use the constant acceleration equations. Part (b) was answered correctly by the vast majority of the candidates.